Claire W. Kim

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EDUCATION

Cornell University | College of Engineering, Sibley School of Mechanical and Aerospace Engineering **Ithaca, NY**
B.S. in Mechanical Engineering *Class of 2028*

SKILLS

SolidWorks, Siemens NX, Fusion 360, MATLAB, Python, GitLab, Linux, OpenAI Codex, KiCAD, Altium, STK, Hypersonics, RF Comms, Space System & Mission Design, Design Synthesis, Conceptual Design, Additive Manufacturing, Machining (Mill, Lathe)

PUBLICATIONS

- *Sailing to the Stars ISS Experiment: Results and Conclusions from Light Sail Deployments in Microgravity.* Small Satellite Conference. Salt Lake City, UT, USA, August 2026.
- *Sailing to the Stars: Free-Flying Light Sails in Microgravity*, Small Satellite Conference. Salt Lake City, UT, USA, August 2025.
- *Unfurling Light Sails for Advanced Deployment Technology: Sailing to the Stars Mission Mechanical Design*, International Symposium on Space Sailing. Delft University of Technology, Delft, The Netherlands, July 2025.

TECHNICAL EXPERIENCE

Space Systems Design Studio (SSDS), Cornell University **Ithaca, NY**
Alpha II, Mission Lead, Mechanical Lead – Undergraduate Researcher (paid) *January 2026 – Present*
- Lead mission development, constructing CONOPS, mission goals, and requirements for Mission Concept Review (May 2026)
- Derive mechanical requirements and research initial concepts for a fixed light sail deployment

DeSCENT, Mechanical Subteam – Undergraduate Researcher (3 credits) *January 2026 – Present*
- Designed, assembled, and integrated a dual-deploy, passive spring-loaded deployer of 2–5 ChipSats at rocket apogee
- Developing an active-release mechanism for a 10-20 ChipSat deployer

Alpha CubeSat, Mechanical Subteam – Undergraduate Researcher (3 credits) *September 2025 – December 2025*
- Developed a deployer and light sail for high-altitude balloon launch and deployment
- Assembled and integrated mechanical and electrical components and systems

Sailing to the Stars, Mechanical Subteam – Undergraduate Researcher (3 credits) *September 2024 – September 2025*
- Developed and tested deployers for light sails with shape-memory alloy frames to understand how nitinol wireframes affect attitude kinematics during light sail deployment in microgravity (aboard the ISS in 2025)
- Iterated CAD design concepts for deployment mechanisms & spacecraft structures through rapid prototyping & verification testing

US Naval Research Laboratory, Naval Center for Space Technology (NCST) **Washington, DC**
Spacecraft Mechanisms Intern, SMART Scholar *Summer 2026*
- Generated mechanism design concepts for a hypersonic flight vehicle control surfaces and developed lofted CAD models to support vehicle geometry and component design
- Designed and 3D-printed a custom encoder housing; developed wiring diagrams and test procedures, executed validation testing, and performed error analysis

Joint Hypersonics Transition Office (JHTO) Intern, Engineering Technician Student Trainee *Summer 2025*
- Diagnosed hypersonic test instrumentation & data acquisition anomalies, identified root causes, & implemented corrective solutions
- Developed and executed component verification and validation tests to evaluate system performance and functionality

Mechanical Lead, Engineering Technician Student Trainee *Summer 2023, Summer 2024*
- Collaborated with a team of 7 HS/college students to develop a 3U CubeSat mission with a radio frequency (RF) payload
- Mechanical Lead - created full-scale mock-ups of CubeSat deployables, designed CAD models & deployable solar/antenna systems

Cornell Nexus Project Team **Ithaca, NY**
Drivetrain & Vibration Feeder, Mechanical Subteam *September 2025 – May 2026*
- Developed and implemented design modifications to resolve anomalies in the current Drivetrain system
- Improved existing Vibration Feeder design and for system integration and testing

AWARDS & HONORS

DoD SMART Scholar **August 2025 – May 2028**
Department of Defense merit-based scholarship for service program.

Girl Scout Gold Award – "Girls in STEM" **September 2023**
Highest award in Girl Scouts, awarded to Scouts who complete an 80-hour minimum global service project.

Crittenberger Award for Community Action and Sports Excellence – Langley HS **May 2024**
Awarded to 2 student-athletes who demonstrate excellence in academics and athletics, and have a significant community impact.

EXTRACURRICULARS & LANGUAGES

SSDS, Society of Women Engineers, Cornell Fashion Collective, Korean-American Student Association
Languages: English (Native), Japanese (Limited Working Proficiency), Korean (Elementary Proficiency)